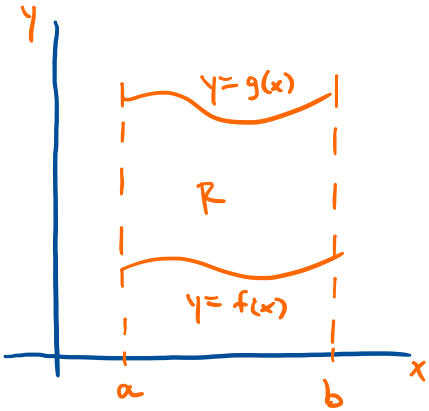
16.2 Iterated integrals

Iterated integrals over rectangles: Suppose $f(x,y)$ is continuous over $R = [a,b] \times [c,d]$. Then

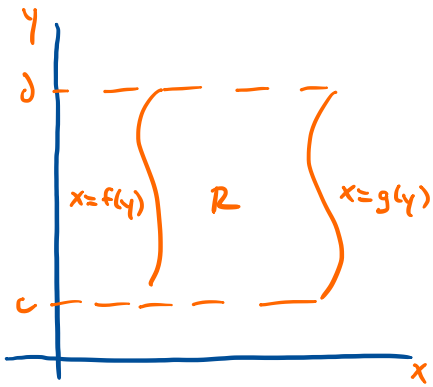
$$\int_R f \, dA = \int_a^b \int_c^d f(x,y) \, dy \, dx = \int_c^d \int_a^b f(x,y) \, dx \, dy$$

→ Do problems 1 and 2.

Iterated integrals over general regions:



$$\int_R f \, dA = \int_{x=a}^{x=b} \int_{y=f(x)}^{y=g(x)} f(x,y) \, dy \, dx$$



$$\int_R f \, dA = \int_{y=c}^{y=d} \int_{x=f(y)}^{x=g(y)} f(x,y) \, dx \, dy$$

Important: The outermost limits must be constants.

→ Do problems 3-5.